

Exposure and Residue Correspondence: Two Silent Failure Modes When Comparing AlphaFold Database Models with Experimental Structures

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Abstract

Comparisons of AlphaFold Database (AFDB) models with experimental structures underpin knowledge of pLDDT, AlphaFold’s prediction of the local distance difference test (IDDT). Two problems can distort them silently: data leakage, when the model has seen the reference or a close relative, and residues paired with the wrong reference atoms. In the ATLAS molecular-dynamics corpus, 791 of 868 proteins (91%) had only references dated on or before AlphaFold2’s training cutoff. In the first four batches of a preregistered census of structures released after 2026-09-11, 87 of 138 proteins (63%) had a same-protein or $\geq 95\%$ -identical structure from before the cutoff, 10 (7%) had no relative at $\geq 30\%$ identity, and 45 models (33%) were given a same-protein template. In an exploratory comparison on an earlier census, proteins with close relatives had slightly higher IDDT, higher pLDDT by a larger margin and similar shares of confident errors. The second problem arose in our pipeline, which applied PDBe/SIFTS `label_seq_id` positions to atoms indexed by `auth_seq_id`. Correcting it changed the median IDDT of 188 of 488 chain segments (151 by more than 0.2) and removed spurious confident errors mimicking AlphaFold overconfidence. A preregistered coverage prediction that tests the IDDT values and pipeline, not pLDDT, failed on the original values; we withdraw that recorded outcome as invalid. A re-evaluation registered after the error was found and a preliminary result was known meets it: mean coverage 0.8945 and 0.9456, slightly below nominal but above the thresholds (nominal minus three Monte Carlo standard errors) of 0.8914 and 0.9449.

1 Introduction

AlphaFold2 [Jumper et al., 2021] and the AlphaFold Protein Structure Database (AFDB; Varadi et al., 2022, Bertoni et al., 2026) have made predicted structures available for hundreds of millions of proteins. For each residue, AlphaFold reports a confidence score that predicts the local distance difference test (IDDT), a superposition-free score of local agreement between a model and a reference structure [Mariani et al., 2013]. The score is called predicted IDDT (pLDDT), and much of what is known about it comes from comparing AFDB models with other data on the same protein. Comparisons with experimental structures, simulations and disorder annotations ask how well pLDDT tracks local error [Jumper et al., 2021], whether it reflects flexibility or disorder [Akdel et al., 2022, Ruff and Pappu, 2021, Guo et al., 2022, Saldaño et al., 2022, Carugo, 2023], and whether conformal intervals built on pLDDT reach their nominal coverage [Shihab et al., 2026]. Two problems can make such comparisons misleading without stopping the pipeline. Agreement can look too good because the model has seen the reference or something close to it. It can look too bad because model residues were paired with the wrong reference atoms.

The first problem is *exposure*, any route by which the reference or a close relative could have informed the model: as a training example, as a *template* (a known structure given to the network as input at prediction time), or as a close homologue in the training data. For some questions, such as describing AFDB as it is used, exposure does not matter. For others, such as accuracy on new proteins or the calibration of pLDDT on unseen folds, agreement with an exposed reference may reflect recall rather than generalisation. This

is data leakage [Kaufman et al., 2012, Kapoor and Narayanan, 2023], and we call the routes by which it can enter an AFDB comparison *exposure channels*. Structure prediction has long guarded against leakage through blind and time-split evaluation [Kryshtafovych et al., 2021, Haas et al., 2018, Jumper et al., 2021], and recent benchmarks show that release-date splits alone still leak through homology [Buttenschoen et al., 2024, Durairaj et al., 2024, Dobson et al., 2025, Bushuiev et al., 2024]. In a retrospective AFDB comparison the model is fixed and part of what it saw is recorded in its model file, so exposure can be measured rather than assumed.

The second problem is *residue correspondence*. Model residues are numbered by their position in the UniProt sequence. Reference atoms are numbered within an entry of the Protein Data Bank (PDB), whose mmCIF (macromolecular Crystallographic Information File) format carries two schemes: a sequential *label numbering* (`label_seq_id`) and an *author numbering* (`auth_seq_id`) chosen by the depositors [Westbrook et al., 2022, RCSB Protein Data Bank, 2026a, wwPDB, 2026]. Resources such as SIFTS (Structure Integration with Function, Taxonomy and Sequences), distributed by the Protein Data Bank in Europe (PDBe), map these numberings to UniProt (Section 2). Less attention has gone to what happens downstream when a pipeline looks atoms up in one scheme using positions given in the other. Forensic re-analyses in high-throughput biology have found that the most common errors are simple ones, such as row or column offsets [Baggerly and Coombes, 2009]. A shifted correspondence produces numbers that look like results.

We measure the first problem in two corpora and report an instance of the second from our own pipeline. Our contributions are the following.

1. A decomposition of exposure into training eligibility, template exposure and homology exposure, with a measurement for each. For AFDB models the template list can be read from the model file (Section 3).
2. In ATLAS [Vander Meersche et al., 2024], a molecular-dynamics corpus that has been reused to train and evaluate structure-ensemble generators [Jing et al., 2024], 791 of 868 proteins have only references released on or before AlphaFold2’s training cutoff (Section 5).
3. We registered a census of PDB entries released after 2026-09-11 before any data existed. In its first four batches (an empty 48-hour batch followed by three weekly ones), 87 of 138 proteins already had a structure of the same protein or a $\geq 95\%$ -identical one in the *training-era PDB* (entries released on or before the 2018-04-30 cutoff), and 45 of the 138 AFDB models were given a same-protein template. In an earlier census, proteins with a close relative had slightly higher IDDT, higher pLDDT by a larger margin and a similar share of confident errors (Section 6).
4. In our census pipeline, a fallback between the two numbering schemes shifted the pairing of whole runs of residues and created residues with high pLDDT and low IDDT. A preregistered coverage prediction, which does not use pLDDT, failed on these values, and we withdraw that recorded outcome as invalid. A re-evaluation registered after the error was found, and after a preliminary result was known, meets the prediction on corrected values, at the 0.95 level by a margin of 0.0007. We give the mechanism, how often it is triggered and the checks that catch it (Section 7, Figure 1).
5. A table of practical checks (Section 8) and a public release of the registrations, code and derived tables.

Sections 2–4 give related work, the exposure channels and the methods, Sections 5–7 the results, and Sections 8–9 recommendations and limitations.

2 Related work

Leakage in machine learning for science. Leakage is a recurring cause of irreproducible results in machine-learning-based science [Kaufman et al., 2012, Kapoor and Narayanan, 2023]. Biology has its own guidance on validation and data splitting [Walsh et al., 2021, Whalen et al., 2022] and tools for splitting biomolecular data by similarity [Joeres et al., 2025]. In structural biology, standard train–test splits leak through sequence and structural similarity in protein–ligand affinity prediction [Li and Yang, 2017, Volkov et al., 2022, Li et al., 2026, Graber et al., 2025], and date-based or metadata-based splits leave similar overlap in docking [Buttenschoen et al., 2024] and protein–protein interaction benchmarks [Bushuiev et al., 2024, Kovtun et al., 2024]. Škrinjar et al. [2026] report that protein–ligand co-folding methods “largely memorize ligand poses from their training data”.

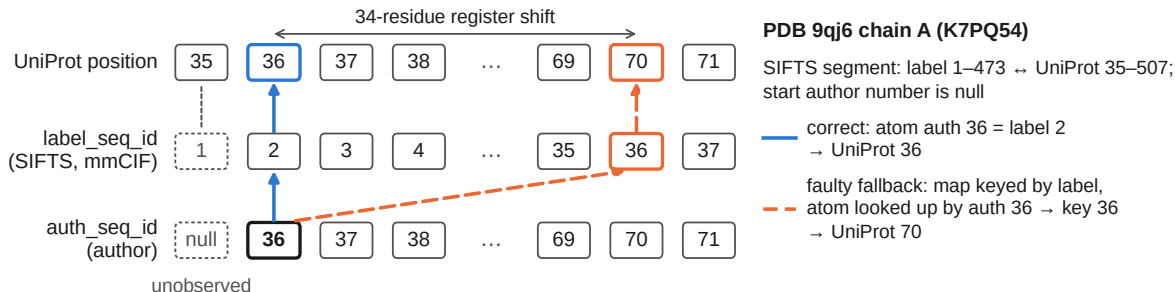


Figure 1: How the numbering fallback shifted PDB entry 9qj6, chain A (protein K7PQ54). Its SIFTS segment (Section 4) begins at an unobserved residue (dashed box), so its starting author number is missing and our code built the residue map from label numbers. It then looked atoms up by author number, which runs 34 ahead of label numbering, so the atom with author number 36 was paired with UniProt position 70 instead of 36. Under this shift the chain’s median IDDT was 0.198 at a median pLDDT of 98.4. With the mapping fixed it is 0.993.

Exposure in AlphaFold evaluation. The AlphaFold2 paper evaluated the model on PDB chains released after its training cutoff and examined template similarity [Jumper et al., 2021]. BETA provides version-specific benchmark sets with sequence and structure filters [Dobson et al., 2025]. Terwilliger et al. [2024] compared AlphaFold predictions made without PDB templates with new crystal structures. AlphaFold’s predictions of fold-switched conformations have been traced to memorised training structures [Chakravarty et al., 2024], and retraining with parts of fold space withheld shows that generalisation survives large omissions but varies with what is withheld [Ahdritz et al., 2024].

Residue correspondence. The Worldwide PDB (wwPDB) remediated the archive to address inconsistencies introduced over the years [Henrick et al., 2008]. Mapping PDB chains to UniProt has long been non-trivial [Martin, 2005]. SIFTS provides residue-level mappings between the two [Velankar et al., 2013, Dana et al., 2019], records them in label numbering and documents the complications added by missing residues, insertions, expression tags and linkers [Choudhary et al., 2023]. Standard sequence alignments often misalign the regions next to unobserved residues [Velankar et al., 2013]. PDBenum rennumbers PDB files to match UniProt [Faezov and Dunbrack, 2021]. Where these resources address how to map residues correctly, we document what one mapping mistake does to a downstream analysis.

Conformal prediction for structure. Split conformal prediction gives finite-sample marginal coverage under exchangeability [Vovk et al., 2005, Lei et al., 2018, Angelopoulos and Bates, 2023]. Residues of one protein are not exchangeable with residues of other proteins, and such hierarchical data need their own guarantees [Dunn et al., 2023, Lee et al., 2026], as do non-exchangeable data more generally [Barber et al., 2023]. CalPro applies evidential and conformal methods to pLDDT [Shihab et al., 2026]. We do not re-evaluate CalPro here.

3 Exposure channels

Let R be an experimental reference structure, first released on date r , compared with an AFDB model M of the same protein. We distinguish three channels through which R , or structures like it, could have informed M . All 138 model files in the first four batches of the preregistered census are AlphaFold2 models (“AlphaFold Monomer v2.0”). AFDB also serves ColabFold models [Mirdita et al., 2022]. Of the 93 model files downloaded for the post-2022 census (Section 4), 6 are ColabFold Monomer v1.5.2 models and 87 are AlphaFold Monomer v2.0, and none of the six belongs to a protein in the exposure comparison of Section 6.3.

Training. AlphaFold2 was trained on PDB structures with a maximum release date of 2018-04-30 [Jumper et al., 2021]. If r is on or before that date, R is *training-eligible*: it may have been a training example. Training

also filtered chains and sampled them by cluster, so eligibility is an upper bound on use of the PDB itself. Self-distillation on predicted structures is a further route that release dates do not capture (Section 9).

Templates. At inference time, AlphaFold2 searches the PDB for templates. The AlphaFold human-proteome study reports the snapshots it used of the PDB and of PDB70, a clustered subset of the PDB used for template search [Tunyasuvunakool et al., 2021], and AFDB states that templates were restricted to structures released before 2021-02-15 [EMBL-EBI AlphaFold Database, 2026]. AFDB model files record the PDB entries supplied as templates in the ModelCIF category `_ma_template_ref_db_details` [EMBL-EBI AlphaFold Database, 2026, Vallat et al., 2023], so this channel can be measured for each model. The list records which templates the network was given, not how much the prediction relied on them.

Homology. Even when R post-dates both cutoffs, the same protein or a close relative may have been in the training-era PDB. This channel is measured by sequence similarity to the training-era PDB [Jumper et al., 2021, Buttenschoen et al., 2024], which release dates cannot capture.

AFDB entries also carry a `modelCreatedDate`, documented as the date the entry was created [Google DeepMind, 2026]. In version 6, unchanged models received new metadata and the v6 label while their coordinates were carried over from version 4 [PDBe, 2025]. The creation date therefore records neither when a prediction ran nor what it could see, and we use it only to exclude models created on or after the census start date.

4 Data and methods

Terms. We use *protein* for a UniProt accession, and call two structures *same-protein* if they map to the same accession. An *entry* is a PDB entry, which may contain several chains. A *chain segment* is the set of residues of one PDB chain mapped to one accession. SIFTS describes it as one or more *SIFTS segments*, each a contiguous stretch with its own start and end. A residue is a *confident error* if its pLDDT is at least 90 and its IDDT below 0.60. At the chain level we use a separate *low-agreement screen*, which flags a chain segment of at least 30 residues with median pLDDT ≥ 90 and median IDDT < 0.4 . A *registration* is an analysis plan, with its code and predictions, frozen in the project repository and timestamped with OpenTimestamps [OpenTimestamps, 2026], which anchors a file’s SHA-256 hash in the Bitcoin blockchain. We call an analysis *preregistered* if it was registered before any IDDT value it uses existed.

ATLAS. When we queried the ATLAS web resource [DSIMB, 2026] on 2026-09-04, it listed 1,938 entry-chain records covering 1,735 PDB entries and 943 proteins, more than the datasets described in the ATLAS paper [Vander Meersche et al., 2024]. Of these proteins, 869 have both an AFDB model and a SIFTS mapping [Velankar et al., 2013, Dana et al., 2019]. One has an AFDB record we could not parse, which leaves 868 proteins and 935 accession-entry pairs. For each reference we took the initial release date, the first record in the mmCIF category `_pdbx_audit_revision_history` [wwPDB, 2026].

Preregistered census. Before any IDDT value existed, we registered a census design, its analysis code and its predictions. The design fixes an AFDB version 6 snapshot at $T_0 = 2026-09-11$ 00:00 UTC and collects, in weekly batches, X-ray entries released after T_0 at 2.5 Å resolution or better. Models created on or after T_0 are excluded. Apart from the release-date filter, which places every reference after AFDB’s 2021-02-15 template cutoff, the design adds no exposure control. What it adds is that every choice in the original design, including two dated amendments made before any data arrived, was fixed before any IDDT value was computed. The residue mapping was replaced later, after the error described in Section 7. We report the first four of 12 planned batches, ending on 4 October 2026: an empty first batch, shortened to 48 hours by one of the amendments, followed by three weekly ones. With the original mapping the four batches gave 488 chain segments from 272 accession-entry pairs and 138 proteins, and with the corrected mapping 495 chain segments and 141 proteins (Section 7.2).

Post-2022 census. An earlier, exploratory census of 168 X-ray entries released after 2022-06-02 used the same author-number fallback as the preregistered census but discarded pairs whose amino acids differed. We use it to check whether findings recur. It was affected by the same error. With the corrected mapping, 139 of its entries from 85 proteins are labelled (96 entries from 64 proteins before), its homology search covered 86 proteins, and 93 AFDB model files were downloaded for it. All its results outside Section 7.4 use the corrected IDDT values.

Residue pairing and IDDT. For each reference chain we pair C α atoms with AFDB model residues through the PDBe SIFTS UniProt mapping. We describe the corrected mapping here, and Section 7 the original and its error. The corrected code (i) indexes the mapping by label numbering (`label_seq_id`) and the chain’s label identifier (`struct_asym_id`), (ii) maps equal-length SIFTS segments by offset and others by global alignment of the chain’s full deposited sequence to the UniProt sequence, (iii) drops any pair whose amino acids differ and (iv) rejects a SIFTS segment if more than 10% of its pairs differ. Chains with fewer than 30 aligned residues are excluded, as are atoms with an alternative location other than A and modified residues other than selenomethionine (16 and 57 residues of the corrected census). We score each residue by IDDT-C α [Mariani et al., 2013], computed per chain against the AFDB model over aligned residues with a 15 Å inclusion radius and distance-difference thresholds of 0.5, 1, 2 and 4 Å. We call the computation of these per-residue IDDT values *labelling*, and computing them again from stored inputs *relabelling*. pLDDT is read from the model’s B-factor column and is never used in labelling.

Homology exposure. For each census protein we queried the search service of the RCSB PDB (Research Collaboratory for Structural Bioinformatics; Burley et al., 2023) for training-era entries in three classes: (i) same-protein entries, (ii) entries containing a chain of a different accession at $\geq 95\%$ sequence identity and (iii) entries containing one at $\geq 30\%$ identity. Sequence searches used MMseqs2 [Steinegger and Söding, 2017] with an E-value of at most 0.1 and the canonical UniProt sequence as query. We set no coverage threshold, so a hit, even a same-protein hit, may cover only part of the protein. Each protein is assigned its strongest class. A failed query would leave a protein unresolved rather than novel. Across the 224 proteins queried for the two censuses, 594 lookups returned results (HTTP 200, two of them repeats of an identical query), 408 ran and found nothing (HTTP 204) and none failed [RCSB Protein Data Bank, 2026b]. Every request and response was cached and hashed.

Template exposure. From each AFDB model file we read the template PDB IDs, mapped each template to UniProt through SIFTS and recorded its release date. A model has a *same-protein template* if any template maps to its accession.

Preregistered coverage prediction. The census registration includes a conformal-coverage prediction that does not use pLDDT. Section 7.5 describes it together with its result.

5 Most ATLAS references predate the training cutoff

For ATLAS we measured only the training channel. We count a protein as training-eligible when all of its references were released on or before 2018-04-30. Of 868 ATLAS proteins, 791 (91.1%) are training-eligible in this sense, and so are 851 of the 935 accession–entry pairs (91.0%). The references were released between 1988 and 2023 (median 2008). Only 77 proteins have any reference released after the cutoff, so agreement between AFDB models and ATLAS references cannot by itself be read as evidence of generalisation. Using the AFDB creation date as the cutoff instead, a tempting but wrong proxy (Section 3), would place every reference before the model for 863 of 868 proteins (99.4%).

ATLAS was built to describe dynamics, and reusing it to study AlphaFold error requires an explicit exposure control. The obvious control is to keep only references released after the cutoff. Section 6 asks how much that achieves.

Table 1: Of the 138 proteins in the first four batches of the preregistered census, 63% had a same-protein or $\geq 95\%$ -identical structure in the training-era PDB and 33% were given a same-protein template. Homology: class of the strongest relative, one per protein. Templates: as listed in the AFDB model file. The three proteins added by the corrected mapping (Section 7.2) are excluded.

Class	Proteins	Share
<i>Homology (training-era PDB)</i>		
Same protein (UniProt accession)	78	56.5%
Different accession, $\geq 95\%$ identity	9	6.5%
Different accession, $\geq 30\%$ identity	41	29.7%
No relative at $\geq 30\%$ identity	10	7.2%
<i>Templates (AFDB model file)</i>		
Same-protein template	45	32.6%
Other templates only	88	63.8%
Unresolved ^a	5	3.6%

^aThe SIFTS lookup failed for at least one template, and no other template maps to the protein.

6 A post-cutoff release does not imply an unseen protein

6.1 Homology exposure

Every structure in the census was released more than eight years after the training cutoff, yet 87 of the 138 proteins (63%) had a same-protein or near-identical ($\geq 95\%$ identity) structure in the training-era PDB, 128 (93%) had a relative at $\geq 30\%$ identity, and only 10 (7%) had none (Table 1). For same-protein hits, the earlier structure may cover only part of the protein. Some well-studied proteins have hundreds of such structures: hen egg-white lysozyme (P00698) had 730 same-protein entries in the training-era PDB, human carbonic anhydrase 2 (P00918) had 700 and β_2 -microglobulin (P61769) had 693. These hits show only that a related structure existed, not whether it differs by ligand, mutation, construct or crystal form.

6.2 Template exposure

The template lists give a more direct measure. All 138 model files list four PDB templates each, released between 1988 and January 2021, so none post-dates the documented 2021-02-15 limit. For 45 proteins (33%) at least one template is a structure of the same protein, for 88 all templates are of other proteins, and 5 are unresolved (Table 1). The homology and template channels overlap only in part. Of the 78 proteins with a same-protein structure in the training-era PDB, 43 were given a same-protein template and 35 were not. Conversely, two proteins with no same-protein structure in the training-era PDB were given a same-protein template (released in 2019 and 2020). One of them, Q9F0J8, has no training-era relative at $\geq 30\%$ identity.

6.3 Is exposure associated with agreement?

To ask whether exposure goes with better agreement, we compared proteins of the post-2022 census with and without a close relative in the training-era PDB (same protein or $\geq 95\%$ identity), using lDDT values recomputed with the corrected mapping (Section 7.4). The comparison was registered in advance as analysis 6 of a separate verification plan in the repository. For each protein we took the median lDDT and the median pLDDT over its residues, and for each group we then averaged these per-protein medians. The 40 proteins with a close relative had a mean median lDDT of 0.975, against 0.959 for the 45 without (difference 0.016, 95% bootstrap interval 0.003–0.032). Their mean median pLDDT was also higher, 96.1 against 92.3 (difference 3.8, interval 1.7–6.0), a larger gap than in lDDT (3.8 against 1.6 points on a 0–100 scale). The mean per-protein share of confident errors was similar in the two groups (0.12% and 0.13%, difference -0.01 percentage points, interval -0.18 to 0.17), although the interval is too wide to rule out a small difference. We did not run this comparison on the preregistered census, whose registered predictions about pLDDT-conditional coverage have not yet been evaluated.

Table 2: The correction removes every chain segment flagged by the low-agreement screen. First four batches of the preregistered census, with IDDT values from the original and the corrected residue mapping.

	Original	Corrected
Labelled residues	117,905	123,454
Chain segments	488	495
Proteins	138	141
Residues with IDDT < 0.60	31.9%	1.1%
Chain segments flagged by the low-agreement screen ^a	116	0
K7PQ54 chain segments with median IDDT < 0.4	29 of 38	0 of 38

^aAt least 30 residues, median pLDDT $\geq$ 90 and median IDDT < 0.4. The 116 come from 32 proteins.

7 A residue-numbering error created spurious confident errors

The exposure measurements need no residue pairing, but any comparison of agreement does. It needs per-residue IDDT values, and computing them for the census is where our own pipeline failed. After the fourth batch the preregistered coverage prediction, which does not use pLDDT, failed at both levels (Section 7.5). A post hoc check found that one protein, K7PQ54, held 14% of the labelled residues and half of the test residues that fell outside the 0.90 intervals. Tracing it led to a residue-numbering error that, unlike exposure, makes the model look worse than it is. The registration and the stored source files let us confirm the cause and relabel the census.

7.1 Mechanism

In the PDBe SIFTS mapping files we used, each SIFTS segment has a start and an end in label numbering and, when the boundary residue was observed, in author numbering. Our census code built each SIFTS segment’s residue map from its starting author number and fell back to the starting label number when the author number was missing, which happens whenever the first residue of the segment is unobserved. Our code then looked atoms up by author number. When the two schemes differed by an offset, every residue of the segment was paired with the wrong UniProt position, a *register shift*. The code had no amino-acid identity check to catch it.

Figure 1 shows chain A of entry 9qj6, a structure of K7PQ54, dihydrolipoyl dehydrogenase from the lesser grain borer *Rhyzopertha dominica* (507 residues; The UniProt Consortium, 2025). The SIFTS segment starts at label residue 1 (UniProt 35), which is unobserved. Author numbering runs 34 ahead of label numbering, so the atom with author number 36 (label 2, UniProt 36) was assigned UniProt 70. Under this shift chain A had a median IDDT of 0.198 at a median pLDDT of 98.4, and chain B, shifted in the same way, had 0.199. With the mapping fixed, chains A and B aligned over 472 and 471 residues with no sequence mismatch and had median IDDT 0.993 and 0.995.

7.2 Scope

We relabelled the four batches from their stored source files, after checking their hashes, once with each mapping. With the original mapping the relabelling reproduced every stored batch exactly, which confirms that it re-executes the preregistered pipeline. One qualifying entry, 9zxa (protein Q92802, 8 chains), is missing from both sets because the original run never fetched its reference file. Table 2 compares the two sets of IDDT values.

Correcting the mapping changed the median IDDT of 188 of the 488 original chain segments. For 151 of them, from 40 proteins, the change exceeded 0.2, and all 151 moved upward (Figure 2). The other 37 changed by at most 0.056. After correction the low-agreement screen flags no chain segment, and the 13 entries whose chains had differed from each other by 0.4 or more in median IDDT no longer do.

The corrected set is not a superset of the original. It labels 6,465 residue positions that the original did not and drops 916 that it did, a net gain of 5,549, and it adds three proteins and seven chain segments that the original mapping had rejected. A reimplementaion with a separately written parser and IDDT code reproduced the corrected IDDT values of 36 chain segments (26 targeted, 10 random) exactly. A

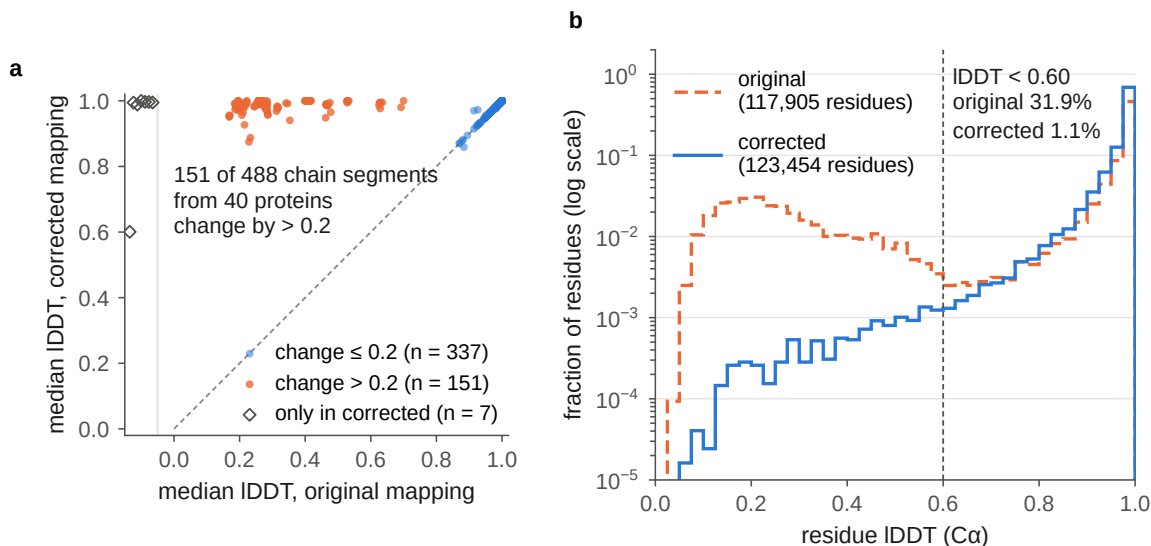


Figure 2: The correction raised every chain segment that it changed by more than 0.2. (a) Median IDDT per chain segment with the original (x) and the corrected (y) mapping. The change is the absolute difference between the two medians, and the dashed line marks no change. The 151 chain segments with a change above 0.2 (orange) moved from original medians of 0.17–0.70 to corrected medians of at least 0.87. Diamonds at the left, spread sideways for visibility, are the seven chain segments that only the corrected mapping labels. (b) Distribution of residue IDDT under each mapping (logarithmic y axis).

second check, with a different mmCIF parser and IDDT implementation [Kunzmann and Hamacher, 2018] and residue-level SIFTS mappings from PDBe, reproduced 494 of the 495 corrected chain segments, every IDDT value within 5×10^{-5} . In the remaining one, chain A of entry 21xg, an alignment tie next to two crosslinked residues makes our corrected mapping pair one residue with the UniProt position next to the one that residue-level SIFTS gives. This moves 18 IDDT values by up to 0.054 and the chain’s median by 0.003, and leaves the share of residues below 0.60 unchanged.

7.3 The error mimics overconfidence, and segment checks catch it

The error created residues with high pLDDT and very low IDDT, the pattern an overconfident model would produce. A calibration study run on these values could have concluded that AlphaFold2 is overconfident on new structures, and our preregistered coverage test did fail on them.

The precondition, a SIFTS segment that begins on an unobserved residue, is common: 304 of the 492 SIFTS segments mapped to the 138 original proteins (62%) begin this way. It causes a shift only when author and label numbering are offset at the start of the segment. That was the case for 147 of the 492 SIFTS segments (30%), from 39 proteins, with offsets from -31 to $+201$. In a comparison we added after the fact, all 147 lay within the 151 chain segments whose median IDDT changed by more than 0.2. Each of the other four of those 151 contains a SIFTS segment whose offset changes partway along it.

Checking amino-acid identity for every pair is necessary but not sufficient. Under a register shift only about 5–7% of pairs match by chance in the segments we tested, and those pairs pass a per-pair filter. What catches the shift is the per-pair check together with rejecting any SIFTS segment in which more than 10% of pairs differ. The post-2022 census had only the per-pair check. It therefore lost most shifted residues but kept those that matched by chance (Section 7.4). Coarser checks catch some cases. A set in which 31.9% of residues fall below 0.60 IDDT is far from the 2.6% seen in the corrected post-2022 census, and 13 entries had chains that differed by 0.4 or more in median IDDT. A chain-agreement check would still miss entries such as 9qj6, where both chains were shifted alike.

Table 3: The preregistered coverage prediction fails on the original IDDT values and holds on the corrected ones. Mean: residue-level test coverage averaged over 200 protein-level splits. Threshold: nominal level minus three MCSE, where the MCSE (Monte Carlo standard error) is the standard deviation of coverage over the splits divided by $\sqrt{200}$. Margin: mean minus threshold.

Nominal	Original IDDT values			Corrected IDDT values		
	Mean	Threshold	Margin	Mean	Threshold	Margin
0.90	0.8580	0.8760	-0.0180	0.8945	0.8914	+0.0031
0.95	0.9152	0.9328	-0.0176	0.9456	0.9449	+0.0007

7.4 The same error in the post-2022 census

In the post-2022 census, 69 entries had been rejected for having too few aligned residues and 3 because the model could not be downloaded. With the corrected mapping, 45 of those 69 entries yield IDDT values. In all, 139 entries are labelled instead of 96 (37,531 labelled residues instead of 24,457). The total is $96 + 45 - 2$ because two previously labelled entries, 10pa and 10ps, are now rejected by the identity checks. The Spearman rank correlation between pLDDT and IDDT rose from 0.617 to 0.643. The pooled residue-level share of confident errors, taken over all residues with pLDDT ≥ 90 , halved from 0.58% ($n = 20,777$) to 0.29% ($n = 31,032$). In the lowest-confidence bin (pLDDT < 50) the share below 0.60 IDDT rose from 58% ($n = 177$) to 75% ($n = 314$).

The added residues and the rise in the lowest-confidence bin come from the 45 newly labelled entries. Most of the rise in rank correlation, and more than half of the fall in confident errors, come from recomputing entries that were already labelled, where shifted residues had matched by chance and passed the per-pair filter. Entry 10ke, for example, held 22 of the 121 original confident errors. Because the September source files had not been kept, this recomputation used files fetched again in October 2026.

7.5 The preregistered coverage prediction before and after correction

The census registration includes a prediction that tests the pipeline rather than AlphaFold2’s confidence. Let y be a residue’s IDDT and $1 - \alpha$ the nominal coverage. For each of 200 random protein-level splits (seeds 0–199), 50% of proteins form a fitting set, 25% a calibration set and 25% a test set. The fitting set serves only to compute $\hat{y}$, a single number: the median IDDT over its residues. Every test residue receives the interval $\hat{y} \pm q$, where q is the $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ -th smallest of the calibration scores $|y - \hat{y}|$ and n_{cal} is the number of calibration residues [Vovk et al., 2005, Lei et al., 2018]. At $1 - \alpha = 0.90$ and 0.95 , we predicted that mean residue-level test coverage over the 200 splits would be at least the nominal level minus three Monte Carlo standard errors (MCSE; Table 3), a threshold that depends on the data. Calibration residues are pooled across proteins of different sizes, and for such hierarchical data marginal residue-level coverage carries no finite-sample guarantee [Dunn et al., 2023, Lee et al., 2026].

On the original IDDT values the prediction failed at both levels (Table 3). Those values were corrupted, so we withdraw that recorded outcome as invalid; a dated correction of record in the repository documents this (Appendix A). The original outputs remain archived unchanged. The original MCSE (0.0080 and 0.0057) was more than twice the corrected one, which lowered the original thresholds; K7PQ54 alone held 14% of the original residues, so whether it fell in the test set could change a split’s coverage considerably. On the corrected values (123,454 residues from 141 proteins) both means are slightly below nominal but pass their thresholds. At 0.90 the mean exceeds the threshold by 0.0031, about one MCSE. At 0.95 it exceeds it by 0.0007, less than half an MCSE (0.0017), so this level passes narrowly and could reverse under a different but equally defensible correction. Counting residues on the interval boundary as uncovered leaves the pass intact (0.9453 against 0.9449).

The seeds and the decision rule were fixed by the original registration, and the corrected mapping and the re-evaluation protocol were registered before the re-evaluation ran. The re-evaluation is not blind. Before registering it, we computed the same statistic outside the registered pipeline on an early version of the corrected mapping (122,024 residues rather than 123,454) and obtained 0.898 and 0.947. The re-evaluation used the registered evaluation code, invoked directly while data collection was paused. Two

Table 4: Checks for AFDB–experiment comparisons and the evidence for each in this paper.

Check	What it catches	How	Evidence here
Training eligibility	Possible training example	Release date on or before 2018-04-30	ATLAS: 791 of 868 proteins (Section 5)
Template exposure	Same protein as template	Template list in the model file, mapped through SIFTS	Census: 45 of 138 models (Section 6)
Homology exposure	Close training-era relative	Sequence search of the training-era PDB	Census: 87 of 138 same-protein or $\geq 95\%$, 10 none at $\geq 30\%$ (Section 6)
Identity check	Register shifts	Amino acids of each pair match; reject segments with $>10\%$ mismatches	Under a register shift only about 5–7% of pairs match by chance in the segments we tested (Section 7.3)
IDDT distribution	Gross corruption	Share of IDDT < 0.60 against a clean set	31.9% original, 1.1% corrected (Section 7.2)
Chain agreement	Shifts in one chain	Median IDDT across an entry’s chains	13 entries flagged, 9qj6 missed (Section 7.3)
Registration, stored inputs	Errors found later	Freeze the design, keep hashed inputs	Relabelling with the original mapping reproduced every stored batch (Section 7.2)

reimplementations written from the registration, without importing the evaluation code, reproduced both means exactly, and one also reproduced the original failure. Together with the mechanism in Section 7.1, this makes the original failure consistent with a labelling artefact. Because the interval ignores pLDDT, the corrected result tests the IDDT values and the pipeline. It says nothing about how well pLDDT tracks error.

8 Recommendations

Table 4 lists the checks that would have exposed the problems reported here. We recommend the following.

1. Report exposure by channel, and state which channels the question requires to be controlled. Release dates alone measure only training eligibility, and database creation dates measure none of the channels (Section 5).
2. For AFDB models, read the template list from the model file (Section 6).
3. Stratify results by sequence identity to the training-era PDB [Buttenschoen et al., 2024], or split data by similarity [Joeres et al., 2025] (Section 6).
4. Pair residues in one numbering scheme, check amino-acid identity for every pair, reject any SIFTS segment with more than 10% mismatches, and run the coarser checks of Table 4 before any downstream analysis (Section 7.3).
5. Treat a failed calibration test as a possible pipeline failure first. Ours is consistent with an error in the IDDT values rather than in the model (Section 7.5).
6. Register analyses before IDDT values exist and keep the raw inputs (Sections 7.2 and 7.4).

9 Limitations

Other AlphaFold versions, including AlphaFold3 [Abramson et al., 2024], have their own cutoffs. We did not measure exposure through multiple sequence alignments or self-distillation, and template lists do not show reliance (Section 3). Homology was measured with fixed identity thresholds, a single canonical sequence and no coverage threshold. A $\geq 30\%$ hit can be a short domain match, which overstates homology exposure, and a single canonical sequence can miss isoform-specific hits, which understates it. IDDT values compare single chains with monomer models and ignore interfaces and ligands in the crystal, which can lower IDDT for reasons unrelated to exposure. We score $C\alpha$ IDDT, so our values are not directly comparable with all-atom IDDT.

The census samples new PDB entries, which over-represent proteins that are crystallised repeatedly, so its exposure shares describe what a post-cutoff release filter yields rather than the novelty of proteins in general. It is reported after four of its 12 batches, and the full census may differ. Because the correction adds and drops residues, the before-and-after coverage comparison is not made on identical residues (Section 7.2). Finally, the numbering error is documented in one pipeline, ours. The trigger frequencies we report describe this census, and we have not examined whether any published analysis is affected.

10 Conclusion

Agreement between an AFDB model and an experimental structure is evidence about AlphaFold only if two things are known: what the model could have seen, and that the residues being compared are the same residues. In ATLAS, 91% of proteins had only training-eligible references. In the preregistered census, whose references all post-date the cutoff, 63% of proteins had a same-protein or $\geq 95\%$ -identical structure in the training-era PDB. In an exploratory comparison, proteins with such a relative had slightly higher IDDT and higher pLDDT. In our own pipeline a numbering fallback, with no segment-level identity check to catch it, made accurate predictions look like confident errors, and a preregistered prediction failed on the resulting values and held once they were corrected. Registration, stored inputs and the checks in Table 4 are cheap enough to be routine for AFDB comparisons.

A Provenance

Table 5: Timeline of the preregistered census and its correction.

Date (2026)	Event
9 Sep	Design, analysis code and predictions registered (proof anchored in Bitcoin 10 Sep)
10 Sep	Two dated amendments before any data: batch 1 shortened to 48 h; repair of the code that dispatches batches
11 Sep	AFDB snapshot time T_0
4–5 Oct	Evaluation after batch 4 records the coverage failure
7 Oct	Audit traces the 9qj6 anomaly to the numbering fallback; collection paused
7 Oct	Unregistered preliminary calculation on an early version of the corrected mapping (0.898, 0.947)
7 Oct	Corrected mapping, re-evaluation protocol and correction of record registered
7 Oct	Batches 1–4 relabelled with the original and corrected mappings; post-2022 census relabelled
7 Oct	Audit of the corrected mapping, the relabelling and the post-2022 recomputation
9 Oct	Registered re-evaluation of the coverage prediction on corrected IDDT values completes
9 Oct	Audits and recomputation of the re-evaluation

In the repository the analyses are numbered e420 (ATLAS), e421 (post-2022 census), e422 (preregistered census) and e427 (corrected mapping and re-evaluation). The correction of record is a dated document, timestamped like the registrations, that withdraws the original outcome of the coverage test and states why. Registrations and intermediate outputs carry timestamped hashes, and the audit reports are in the `audits/` directory.

Data and code availability

The registrations with their timestamp proofs, the analysis code, the derived residue-level tables, the SHA-256 hashes of every request and response, and the audit reports are available at <https://github.com/MachineLearning-Nerd/afdb-exposure-audit>, with a script that checks 208 numbers in the paper, 147 of them recomputed from the row-level tables and the rest reconciled with them or read from stored outputs. The full evaluation records of the preregistered census contain registered predictions that have not yet been evaluated and will be added when the census closes. Raw files from the PDB, AFDB, PDBe/SIFTS, UniProt

and ATLAS are not redistributed but can be re-fetched, and the recorded hashes identify the exact files used. The ATLAS counts in Section 4 (records, entries, and proteins with a model and a mapping) depend on a candidate-pool file from a separate ongoing study that is not released, so that step cannot be rerun from the repository. Its outputs and run records are included.

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